Miguelyser

A simple Atherosclerotic Plaque Analyzer

User Guide

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Automated segmentation and quantification of atherosclerotic plaques in Oil Red O-stained mouse aorta images.

This document covers: installation and setup, a step-by-step guide to using the graphical interface, a detailed explanation of the image analysis algorithm (with pros and cons), and instructions for manual correction when automatic detection is insufficient.

1 Introduction

What is Atherosclerosis?

Atherosclerosis is a chronic inflammatory disease of the arteries in which lipid-rich plaques accumulate within the vessel wall. Over time these plaques grow, harden, and can narrow or block blood flow, leading to heart attacks and strokes. Mouse models — particularly $ApoE^{-/-}$ and $Ldlr^{-/-}$ mice fed a high-fat diet — develop plaques throughout the aorta and are widely used to study disease progression and test new therapies.

Oil Red O Staining

Oil Red O (ORO) is a fat-soluble dye that stains neutral lipids (triglycerides and cholesteryl esters) **bright orange-red**. In the *en face* aorta preparation, the intact aorta is pinned flat on a black wax surface, fixed, and stained. The result is a Y-shaped tissue — grayish-purple aorta wall on a dark background — with orange-red patches wherever atherosclerotic plaques are present. The proportion of the aortic surface covered by plaques (plaque burden) is the primary readout of disease severity.

What Miguelyser Does

Miguelyser loads JPEG images of ORO-stained aortas, automatically detects the aorta and the plaques using colour- and morphology-based image analysis, and divides the Y-shaped aorta into three anatomical regions:

- **Trunk** — the lower vertical section of the aorta.
- **Left arm** — the left iliac branch (upper-left).
- **Right arm** — the right iliac branch (upper-right).

For each region it calculates the aorta area, the plaque area, and the plaque burden (%). Results are displayed in the GUI, saved as annotated images, and exported to an Excel spreadsheet.

2 Installation & Setup

Miguelyser is distributed as a **standalone application** — no Python knowledge or installation is required for end users. Simply download the file for your operating system and double-click it.

Option A — Standalone Application (Recommended)

■ This is the recommended route for most users. You do not need to install Python, pip, or any other software.

Windows

1. Obtain the file **Miguelyser.exe** from whoever sent you this guide.
2. Save it anywhere on your computer (e.g. your Desktop or Documents folder).
3. **Double-click Miguelyser.exe** to launch the application.

■ Windows SmartScreen may show a warning the first time you run the application because it is not yet recognised by Microsoft. Click *More info*, then *Run anyway* to proceed. This is normal for new applications from individual researchers.

macOS

1. Obtain the file **Miguelyser** (no extension) from whoever sent you this guide.
2. Move it to your Applications folder or Desktop.
3. **Double-click Miguelyser** to launch the application.

■ macOS Gatekeeper may show a warning that the app is from an unidentified developer. To open it: right-click (or Control-click) the file and choose *Open*, then click *Open* again in the dialog. You only need to do this the first time.

Option B — Running from Source (Advanced / Developer)

If you prefer to run the Python source code directly, or if you want to modify the software, follow the steps below. This requires Python 3.9 or later.

Required packages

Package	Purpose
opencv-python	Core image processing (thresholding, morphology, contours)
numpy	Array and matrix operations (bundled with OpenCV)
Pillow	Image loading and display inside the GUI
scipy	Median filter used in bifurcation detection
pandas	Tabular data and Excel export
openpyxl	Excel (.xlsx) file writing engine for pandas

Package	Purpose
reportlab	PDF guide generation only — not needed to run the GUI

Step-by-step

1. **Install Python 3.9+** — download from *python.org* or use Miniconda (*docs.conda.io*). Accept the default settings and tick *Add Python to PATH* on Windows.
2. **Open a terminal** — on macOS use *Terminal*; on Windows use *Command Prompt* or *Anaconda Prompt*.
3. **Install the required packages:**

```
pip install opencv-python Pillow scipy pandas openpyxl
```

1. **Place the script** — copy *aorta_analyzer.py* to any folder.
2. **Launch the application:**

```
# macOS / Linux python3 aorta_analyzer.py # Windows python aorta_analyzer.py
```

Building the Standalone Application (for Developers)

The standalone application is built from the Python source using **PyInstaller**. Build scripts are provided alongside the source code.

File	Platform	What it does
build_windows.bat	Windows	Builds Miguelyser.exe in the dist\ folder
build_macos.sh	macOS	Builds the Miguelyser app in the dist/ folder

■ PyInstaller can only build for the platform it runs on — to build a Windows .exe you must run build_windows.bat on a Windows machine, and to build a macOS app you must run build_macos.sh on a Mac.

3 Using the GUI

Startup Splash Screen

When you launch Miguelyser a splash screen appears for approximately 3 seconds before the main window opens. You can click anywhere on it to dismiss it immediately.

Main Window Overview

The main window is divided into five areas:

- **Toolbar** (top, dark bar) — coloured action buttons and a live status message.
- **Progress bar** — appears below the toolbar during batch processing and shows how many images have been completed.
- **Colour legend strip** — a compact reference showing which colour corresponds to which aorta region in the annotated image.
- **Image list** (left panel) — filenames in the selected folder. Images that have been analysed are prefixed with a green ✓.
- **Image panels** (centre) — *Original* on the left, *Annotated* on the right. The total plaque burden percentage is displayed in large text below the annotated panel.
- **Results table** (bottom) — one row per processed image with all 13 measurements. Scrollable both horizontally and vertically.

Toolbar Buttons

Button	Colour	What it does
■ Open Folder	Blue	Opens a folder-chooser dialog. All image files (.jpg, .jpeg, .png, .tif) in the chosen folder are listed in the image list. Annotated images (names ending with _*annotated*) are automatically excluded. The first image is displayed immediately.
■ Process All	Green	Runs the full analysis pipeline on every image in the list. A progress bar tracks completion. Each annotated image is saved alongside its source as *filename_annotated.jpg*. Processing happens in a background thread so the window stays responsive.
■ Process Selected	Teal	Analyses only the currently highlighted image. Useful when you want to re-process a single image after editing its boundaries.
■ Export Excel	Purple	Saves all results to *plaque_results.xlsx* in the selected folder. The file contains one row per image and 13 columns (see Section 4.6). If the file is open in Excel, close it first.
✏ Edit Boundaries	Orange	Opens the manual correction window for the currently selected image (see Section 6).
✕ Cancel	Red	Stops the current batch processing cleanly after the current image finishes. Only enabled during batch processing.

■ Hover the mouse over any button for a short tooltip describing what it does.

Keyboard Navigation

Use the ← **Left** and → **Right** arrow keys to step through images in the list. The original and annotated panels update instantly — no re-processing occurs for images that have already been analysed. This makes it easy to review a whole batch quickly.

Double-click either image panel to open a full-screen zoom view of that image. Press **Escape** or click anywhere to close the zoom view.

Typical Workflow

1. Click ■ **Open Folder** and navigate to the folder containing your aorta images.
2. Click ■ **Process All** and wait for the progress bar to reach 100 % and the status to say *Done*.
3. Press → repeatedly to flip through all images. Double-click the Annotated panel to zoom in and check the overlays.
4. For any image where detection looks wrong, select it in the list and click ⥰ **Edit Boundaries** to correct it (see Section 6).
5. Click ■ **Export Excel** to save the final measurements to a spreadsheet.

Reading the Annotated Image

Colour / Style	Meaning
Solid green outline	Full aorta boundary
Blue dashed outline	Individual plaque contours (Oil Red O staining)
Blue-tinted fill	Trunk region (lower vertical section)
Yellow/cyan fill	Left arm region (upper-left branch)
Magenta fill	Right arm region (upper-right branch)
White circle	Detected bifurcation point (Y-junction)
Text label on each region	Region name and its plaque burden %
Green banner (top-left)	Total plaque burden across the whole aorta

■ Left / right arm labels are from the *image* perspective, not the animal's anatomy. Verify the orientation of your images before interpreting the arm labels biologically.

4 Algorithm Explained

Miguelyser processes each image through six sequential steps. Understanding these steps helps you know which sliders to adjust when automatic detection goes wrong.

4.1 Circular Field Detection

Microscopy images have a dark circular border — the aperture of the lens. The algorithm thresholds the grayscale image at intensity 20 and finds the largest bright contour. This contour defines a **circle mask** that is applied to all subsequent steps, preventing the dark corners from being mistakenly analysed.

4.2 Aorta Segmentation

The aorta is brighter than the dark background. The algorithm uses the following pipeline to isolate it:

1. **Grayscale conversion + Gaussian blur (5×5 kernel)** — converts the colour image to grayscale and smooths it to reduce noise from pins, bubbles, and surface reflections.
2. **Otsu thresholding (modified)** — Otsu's method automatically finds the optimal threshold that separates the bright tissue from the dark background. The computed threshold is multiplied by 0.80 (conservative) and then floored at the *aorta brightness min* parameter (default 55). This balances sensitivity against picking up dim connective tissue.
3. **Morphological opening (default kernel 13 px)** — erosion followed by dilation. Removes thin structures such as pins and bristles that are narrower than the kernel radius.
4. **Largest connected component selection** — the aorta is the biggest bright blob in the image. All smaller blobs (satellite tissue fragments) are discarded at this stage.
5. **Morphological closing (default kernel 27 px)** — dilation followed by erosion, applied only to the aorta body. Fills small holes caused by pins and staining artefacts.
6. **Bridge-breaking** — the large closing kernel can sometimes bridge a thin gap between the aorta and an adjacent tissue fragment. To fix this, the closed mask is eroded by kernel/4 (≈6 px), the largest component is re-selected, and then dilated back. Any bridge narrower than 6 px is severed; the thick aorta body (≥80 px) survives intact.

4.3 Plaque Detection

Oil Red O plaques are orange-red — a colour that is easy to isolate in the **HSV colour space** (Hue, Saturation, Value), because hue encodes colour independently of brightness.

- **Primary range**: Hue 0–22, Saturation ≥ 110, Value ≥ 75 — captures orange through red.
- **Wrap-around range**: Hue 165–180, same S/V bounds — catches pure red (which wraps around in OpenCV's 0–180 H scale).
- **Morphological opening (default 5 px)** — removes isolated colour speckles smaller than the kernel.
- **Aorta mask** — only pixels inside the detected aorta are kept, preventing background staining from being counted.

4.4 Bifurcation Detection

To split the aorta into trunk and arms, the algorithm must find the Y-junction (bifurcation point). It uses a **row-scan strategy**:

1. **Search window** — only the top 65% of the aorta height is scanned. This prevents a tiny artefact gap at the very tip of the trunk from being mistaken for the bifurcation.
2. **Method A (primary)** — scan rows from top to bottom and record the deepest row that contains two or more separate pixel runs with a gap of at least 10 px between them. That row is the bottom edge of the arm region.
3. **Method B (fallback)** — if the closing step has merged the two arms into one blob (no gap visible), the algorithm uses a width-profile approach: the combined arms are much wider than the trunk, so the bifurcation is the row where the per-row pixel count drops below 55% of the maximum arm width (after median smoothing).

4.5 Region Segmentation

Once the bifurcation point (bif_x, bif_y) is found, the aorta mask is split by simple coordinate rules:

- **Trunk**: all aorta pixels where y > bif_y (below the bifurcation).
- **Left arm**: pixels where y ≤ bif_y AND x < bif_x.
- **Right arm**: pixels where y ≤ bif_y AND x ≥ bif_x.

> ■ Left / right are from the image perspective, not the animal's anatomy. Verify the orientation of your images before interpreting the arm labels biologically.

4.6 Area Quantification

Areas are measured in **pixels**. For each region the plaque burden percentage is:

```
plaque_pct = 100 × plaque_area_px / region_area_px
```

The Excel output contains 13 columns: filename, then aorta area, plaque area, and plaque % for the whole aorta, then the same three metrics repeated for trunk, left arm, and right arm.

5 Pros and Cons of the Algorithm

The table below summarises the strengths and weaknesses of each algorithmic step so you can make informed decisions when adjusting parameters or applying manual corrections.

Algorithm step	Advantages	Limitations
Otsu thresholding (aorta)	Adaptive — no single fixed threshold needed; adjusts to the brightness of each image automatically.	Can fail if background illumination is very uneven or if staining intensity varies greatly across the image.
Largest connected component	Reliably isolates the aorta as the dominant bright structure without requiring shape-based rules.	Fails if another tissue fragment is bigger than the aorta, or if the aorta splits into two disconnected pieces in the thresholded image.
Morphological opening / closing	Fills pin holes and removes thin artefacts automatically, reducing the need for manual cleanup.	Large kernels can bridge thin gaps between the aorta and satellite blobs (mitigated by the bridge-breaking step).
Bridge-breaking (erode–reselect–dilate)	Cleanly severs thin morphological bridges while preserving the thick aorta body. No user interaction required in most cases.	If the satellite blob is very close to the aorta and the bridge is wider than kernel/4, it may survive. Requires manual correction.
HSV plaque detection	Hue-based thresholding is robust to moderate brightness variation; the dual-range (0–22 + 165–180) catches all orange/red tones.	Very pale or washed-out plaques may fall below the saturation/value thresholds and be missed. Deep purple haemorrhage may be included.
Row-scan bifurcation	Simple and fast; correctly handles images where the gap between arms is clearly visible after thresholding.	Falls back to the less precise width-profile method when arms are merged by morphological closing. Accuracy depends on image quality.

6 Manual Adjustments — Edit Boundaries

When to Use Edit Boundaries

Automatic detection works well for well-stained, evenly illuminated images. Open the **Edit Boundaries** window when:

- The aorta mask includes satellite tissue blobs.
- Part of the aorta is missing from the mask (under-detected).
- Plaques are missed or the plaque mask includes non-plaque tissue.
- The bifurcation marker is placed in the wrong location.

Select the image in the list, then click **Edit Boundaries**. The correction window opens with a live preview. Changes take effect immediately.

Threshold Sliders

Moving a slider updates the detection algorithm in real time — you see the result within about one second. Start with sliders before resorting to the paint brush, because slider changes are non-destructive and can be reset with **Reset to Auto**.

Slider	Default	Range	Effect & suggestions
Aorta brightness min	55	20 – 150	Sets the minimum grayscale value for a pixel to be included in the aorta. **Increase** (60–90) if dim background tissue is being picked up. **Decrease** (30–50) if the aorta has pale areas that are being missed.
Aorta open kernel	13	3 – 40	Size (px) of the opening kernel that removes thin structures. **Increase** (15–25) to remove thicker pin artefacts. **Decrease** (5–10) if narrow parts of the aorta are being eroded away.
Aorta close kernel	27	5 – 60	Size (px) of the closing kernel that fills holes. **Increase** (30–45) if the aorta has large pin holes. **Decrease** (15–22) if closing is bridging to satellite blobs despite the bridge-breaking step.
Plaque hue max (0–180)	22	5 – 40	Upper hue bound for plaque detection (OpenCV scale 0–180). **Increase** (25–32) to include yellowish-orange plaques. **Decrease** (12–18) to exclude background reddish tissue.
Plaque sat min	110	40 – 200	Minimum saturation. **Decrease** (70–100) to pick up pale, less intensely stained plaques. **Increase** (130–160) to exclude background redness with low saturation.
Plaque val min	75	30 – 180	Minimum brightness (value). **Decrease** (40–60) to capture very dark red plaques. **Increase** (90–120) to exclude dim background staining.
Plaque open kernel	5	1 – 20	Removes colour speckles smaller than this kernel. **Increase** (7–12) to eliminate more noise. **Decrease** (1–3) if small genuine plaques are disappearing.

Paint Brush

When sliders alone cannot fix the problem, use the paint brush to directly add or erase pixels from the aorta or plaque masks. Click and drag on the preview image.

Brush mode	Use case
Aorta ADD	Paint missing aorta area (e.g. a pale region the algorithm missed).
Aorta ERASE	Remove incorrectly included tissue (e.g. a satellite blob that survived bridge-breaking).
Plaque ADD	Mark a plaque the HSV threshold missed. Added pixels are automatically clipped to the aorta boundary.
Plaque ERASE	Remove false-positive plaque detections (e.g. red connective tissue outside the aorta, or tissue at the aorta edge).

- **Brush size**: use a large brush (30–60 px) for coarse corrections and a small brush (5–15 px) for fine detail. Brush radius is shown in image pixels.
- **Reset to Auto**: discards all brush strokes and slider changes, reverting to the original automatic detection.
- **Apply & Save**: recalculates all measurements from the corrected masks, updates the results table, saves the corrected annotated image, and closes the correction window.

Tips for Common Problems

Problem	Suggested fix
Satellite blob included in aorta	First try: decrease *Aorta close kernel*. If the blob persists, use Aorta ERASE brush to paint it out.
Pale aorta region missing	Decrease *Aorta brightness min* by 10–15 units, or use Aorta ADD brush on the missing area.
Plaques not detected	Decrease *Plaque sat min* and/or *Plaque val min*. If plaques are more orange than red, increase *Plaque hue max*.
Too much background counted as plaque	Increase *Plaque sat min*. Use Plaque ERASE brush on the false-positive areas.
Bifurcation marker in wrong position	Adjust *Aorta close kernel* — a larger kernel merges the arms and triggers the width-profile fallback (less accurate). Try a smaller value so the gap between arms is visible to the row-scan. Note: you cannot directly drag the bifurcation point; correct it by adjusting the aorta mask shape.
Arms labelled the wrong way	This is an image orientation issue. Flip the image horizontally in an external editor before loading into Miguelyser.

7 Troubleshooting

Standalone Application Issues

Symptom	Likely cause	Fix
Windows: 'Windows protected your PC' warning	Windows SmartScreen blocks unsigned apps from new developers.	Click *More info*, then *Run anyway*. This is a one-time step and is safe for software from a known source.
macOS: 'cannot be opened because the developer cannot be verified'	Gatekeeper blocks apps not signed with an Apple certificate.	Right-click (Control-click) the app file, choose *Open*, then click *Open* in the dialog. You only need to do this once.
App opens briefly then closes immediately	A missing system library or antivirus interference.	Temporarily disable antivirus and try again. On Windows, check that the Visual C++ Redistributable is installed (available free from Microsoft's website).
App is slow to start (10–30 seconds)	Normal behaviour on first launch — the standalone bundle extracts libraries to a temporary folder.	Wait for the splash screen to appear. Subsequent launches will be faster.

Analysis Issues

Symptom	Likely cause	Fix
Images do not appear after opening folder	Image files may be in an unsupported format, or the folder path contains special characters.	Ensure files end in .jpg, .jpeg, .png, or .tif. Avoid folder names with special or accented characters.
Processing hangs / window unresponsive	Very large images (>10 megapixels) can take several seconds per image.	The progress bar will update once each image finishes. Use ■ Process Selected on one image first to check the speed.
Excel file not saved	No images processed yet, or the folder is read-only, or the file is open in Excel.	Process at least one image first. Close plaque_results.xlsx in Excel if it is already open. Check folder write permissions.
Annotated images not saved	Write permission denied, or disk is full.	Check available disk space and ensure you have write permission to the image folder.

Source / Python Issues

Symptom	Likely cause	Fix
ModuleNotFoundError: No module named 'cv2'	opencv-python not installed in the active Python environment.	Run: pip install opencv-python. Make sure you launch the script with the same Python interpreter that pip used.

Symptom	Likely cause	Fix
ModuleNotFoundError: No module named 'scipy'	scipy not installed.	Run: pip install scipy

■ For persistent source-installation issues, verify your package versions: python3 -c "import cv2, PIL, scipy, pandas; print(cv2.__version__, PIL.__version__, scipy.__version__)"